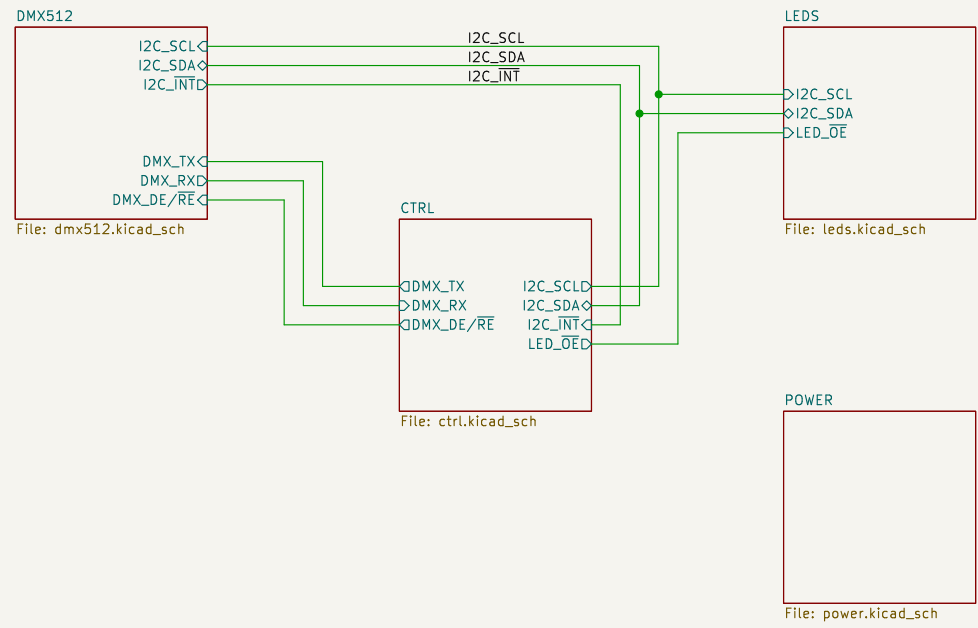


# LED Beacon

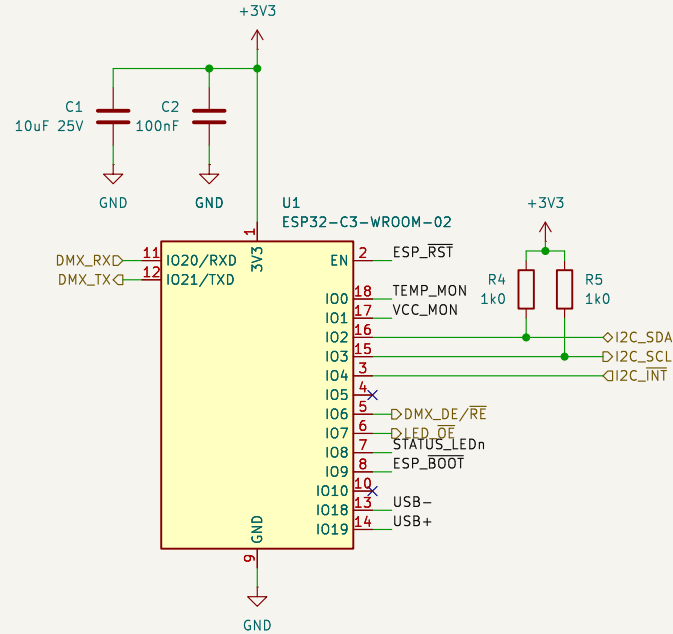
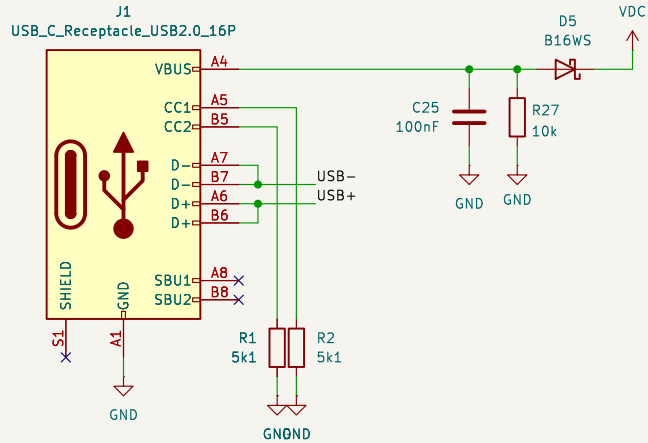


- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

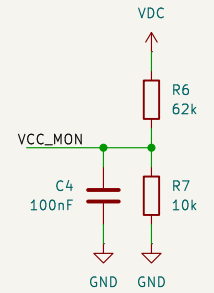
|                            |                  |         |
|----------------------------|------------------|---------|
| <b>Wires &amp; Bits</b>    |                  |         |
| Sheet: /                   |                  |         |
| File: emf_beacon.kicad_sch |                  |         |
| <b>Title:</b>              |                  |         |
| Size: A4                   | Date: 2026-04-10 | Rev: 0  |
| KiCad E.D.A. 9.0.9         |                  | Id: 1/5 |

# Controller

## USB Programming

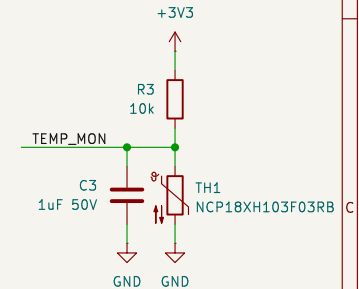


## VCC Monitoring

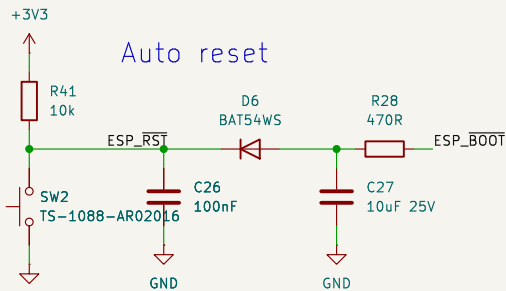


Voltage Monitor  
 $V_{max}$  is 17.5V  
 $V_{adc} \times ((43k\Omega + 10k\Omega) + 10k\Omega) = VCC$   
 $V_{adc} \times 5.3 = VCC$   
 $1lsb = 4.271\mu V$

## Temperature Monitoring

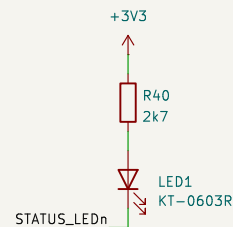


## Auto reset



Short press resets the ESP32  
 Long press resets the ESP32  
 and puts it in bootloader mode

## Status LED



I08 needs to be held high to boot  
 so LED & resistor are high side.  
 Set STATUS\_LEDn to 0V to light LED.

## Wires & Bits

Sheet: /CTRL/  
 File: ctrl.kicad\_sch

## Title:

Size: A4  
 Date: 2026-04-10  
 KiCad E.D.A. 9.0.9

Rev: 0  
 Id: 2/5

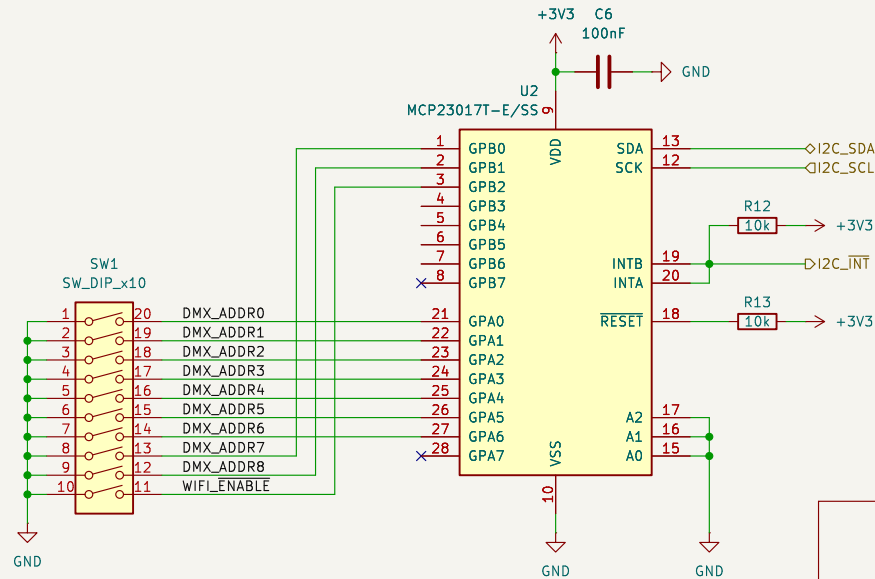
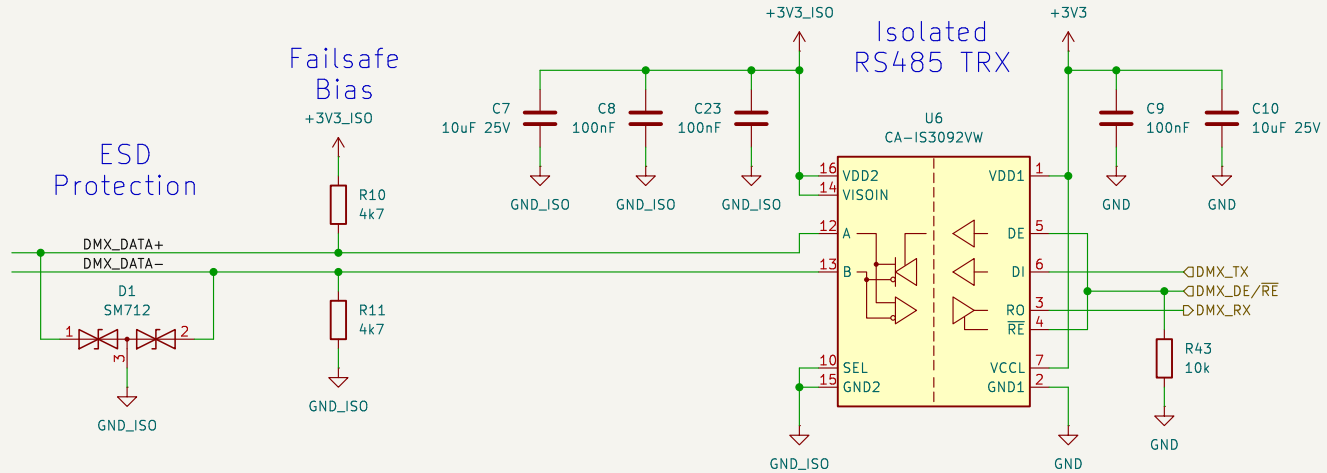
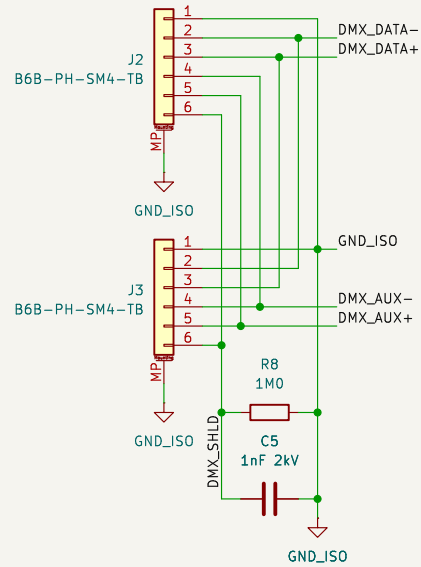
# DMX512

Failsafe Bias  
+3V3\_ISO

ESD Protection

Isolated RS485 TRX

To XLRs



I<sup>2</sup>C Addr: 0x20  
MCP23017 has a fault with GPA7 & GPB7.  
The can corrupt an I<sup>2</sup>C transaction if  
they change state during it.

## Wires & Bits

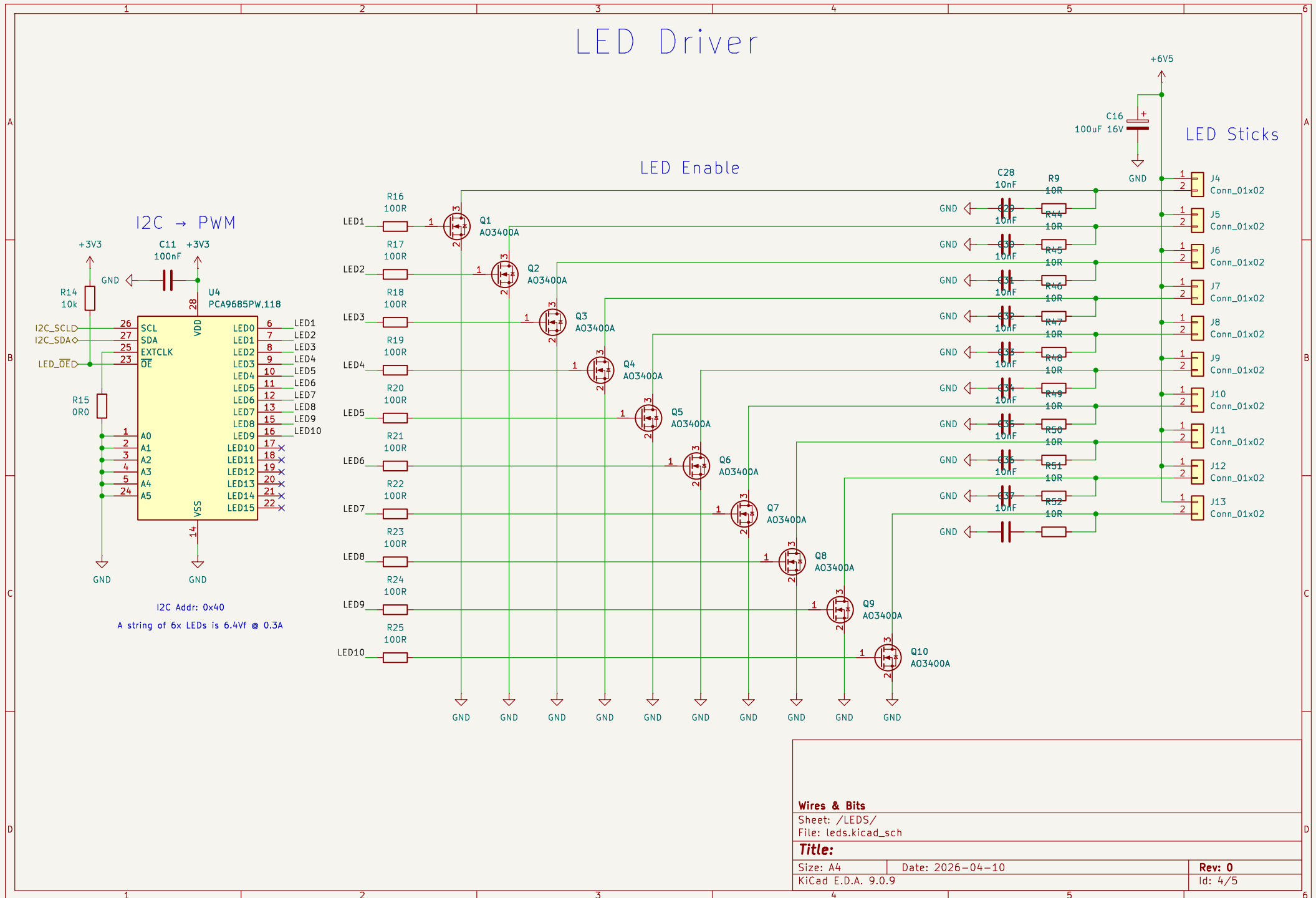
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File: dmx512.kicad\_sch

## Title:

Size: A4  
Date: 2026-04-10  
KiCad E.D.A. 9.0.9

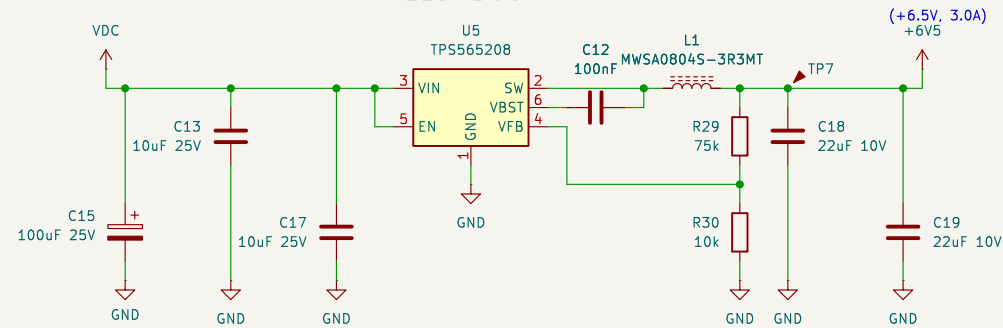
Rev: 0  
Id: 3/5

# LED Driver

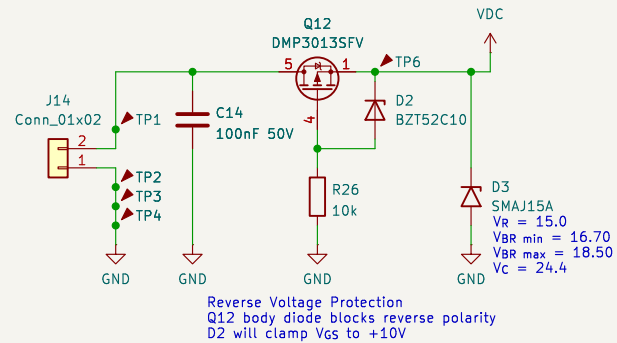
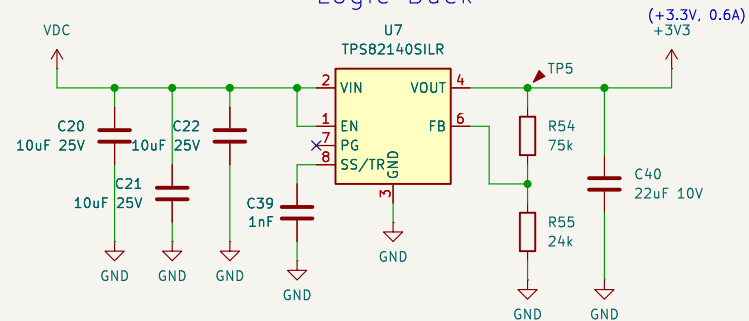


# Power Supplies

## LED Buck



## Logic Buck



EMC ferrites /CMC?

### Wires & Bits

Sheet: /POWER/  
File: power.kicad\_sch

### Title:

Size: A4  
Date: 2026-04-10  
KiCad E.D.A. 9.0.9

Rev: 0  
Id: 5/5